

Validating Multi-Model LLM Ensembles for Survey Harmonization Classification

DRAFT — not validated — see disclaimer slide

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Disclaimer

Warning

DRAFT — These materials are in development and have not been independently validated.

The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.

This is exploratory research assessing feasibility of AI-assisted survey analysis methods.

For survey-specific findings, see Report 3A: Survey Harmonization Findings

Research Question

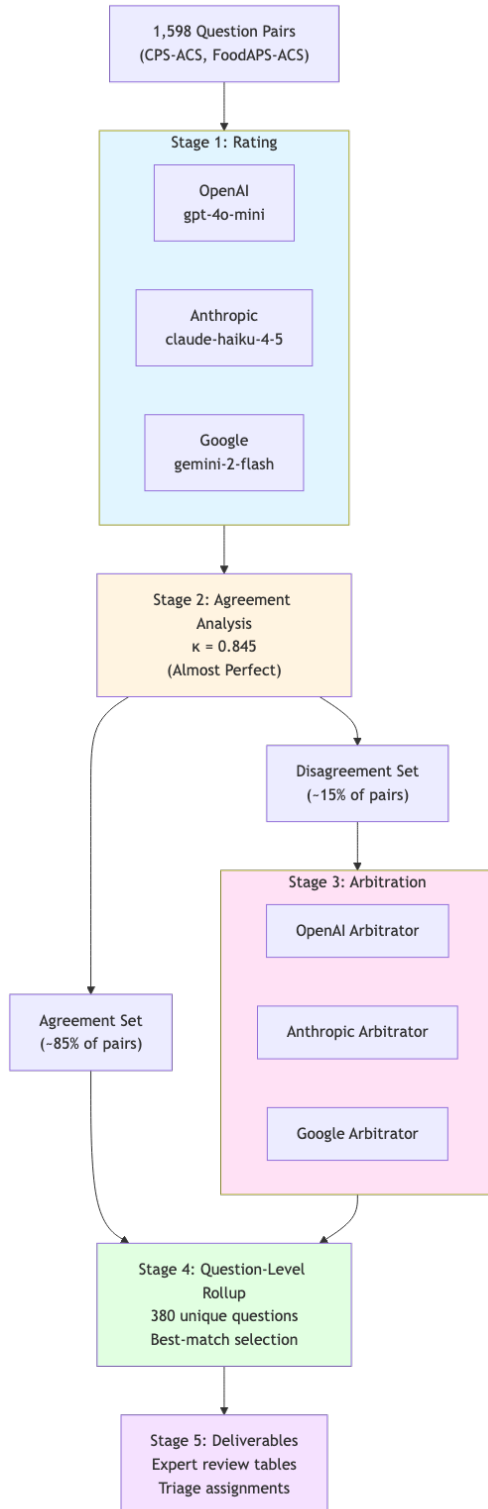
Can multi-model LLM ensemble reliably classify survey question harmonization compatibility?

- **Challenge:** Survey harmonization requires nuanced expert judgment
- **Approach:** Three-tier architecture (raters → arbitrators → triage)
- **Validation:** Construct validity through independent model convergence
- **Analysis:** Behavioral patterns in arbitrator decision-making

Pairwise Comparison Task

- Within concept categories, compare all question pairs
 - **1,598 pairs** analyzed (CPS-ACS, FoodAPS-ACS)
 - Harmonization classification: F1/F2/F3 with barrier codes
 - Naive but exhaustive — expect many F3, looking for matches
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Three-Tier Architecture



1. **Rating tier:** 3 fast models independently classify all pairs
2. **Arbitration tier:** 3 flagship models resolve disagreements
3. **Triage tier:** Best-match rollup with confidence scoring

Multi-Model Ensemble Design

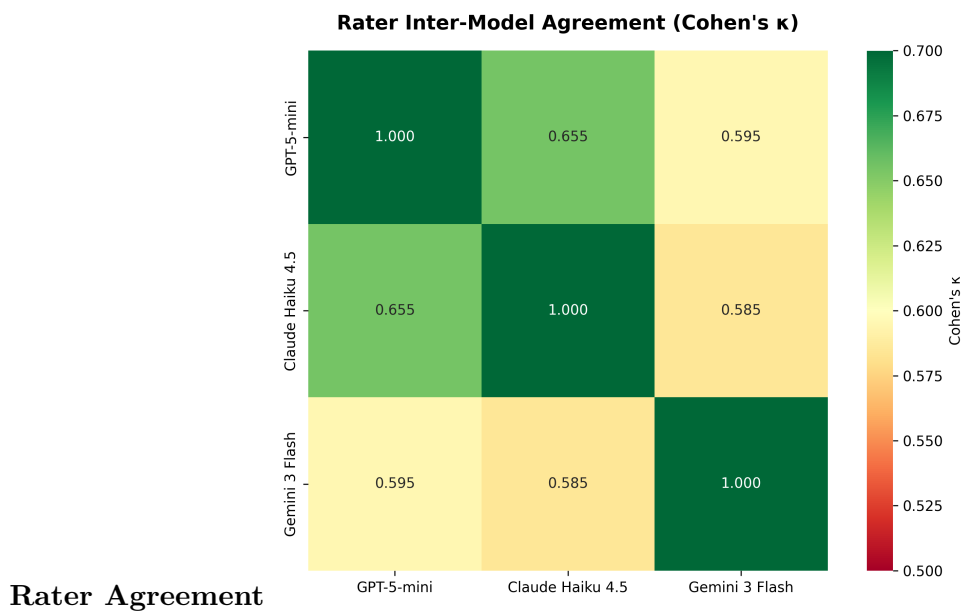
Three frontier models to reduce bias:

Model	Behavior
OpenAI	Moderate synthesis, slight pro-self bias
Anthropic	High synthesis, neutral
Google	Deferential, conservative

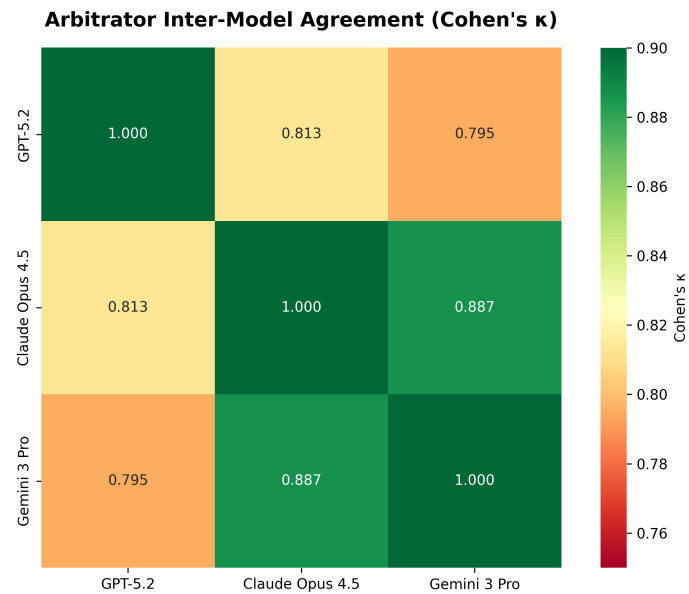
Different profiles = documented finding, not noise

Construct Validity: Why Trust These Results?

Three independent architectures → same judgments = well-defined task



Fleiss κ = 0.611 (substantial)



Arbitrator Agreement

Fleiss κ = 0.843 (almost perfect)

Convergence INCREASES with reasoning depth (0.611 $\rightarrow$ 0.843)

Rater Agreement

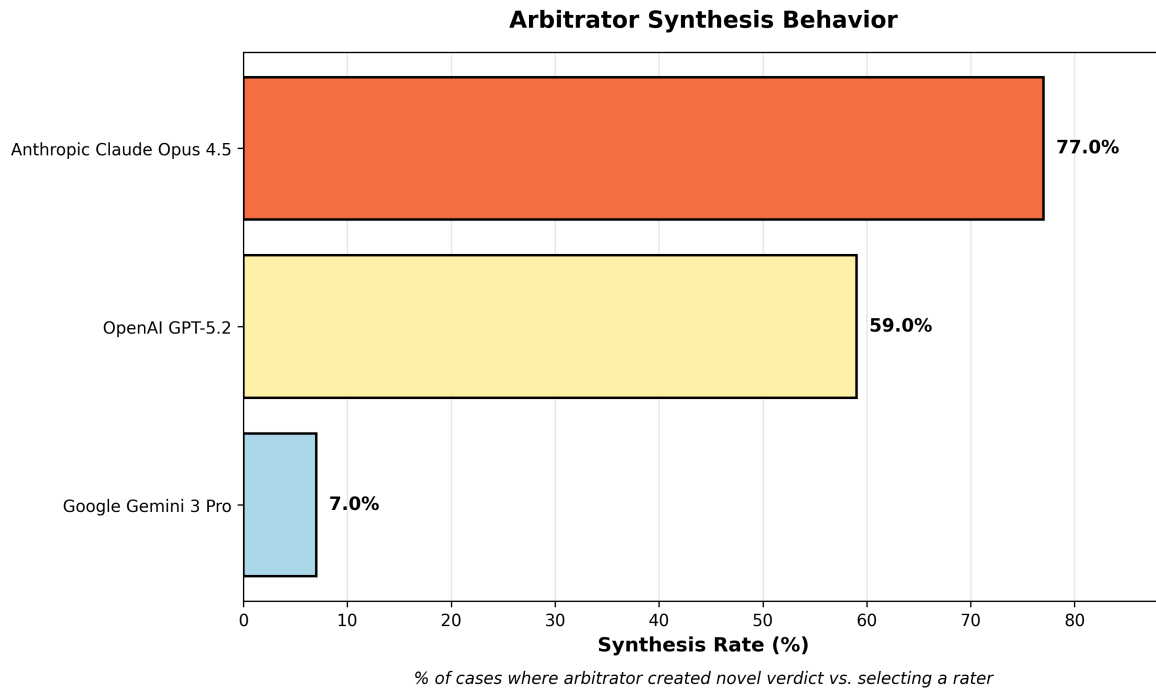
- κ = **0.611** (Fleiss) — substantial agreement
- Three independent architectures converge on same patterns
- Disagreement rate manageable for arbitration tier

Arbitrator Agreement

- κ = **0.843** (Fleiss) — almost perfect agreement
- Flagship models with deeper reasoning $\rightarrow$ higher convergence
- Arbitration tier successfully resolves rater disagreements

Arbitrator Behavioral Differences

Not a bug — a feature. Diversity captures multiple valid interpretations.

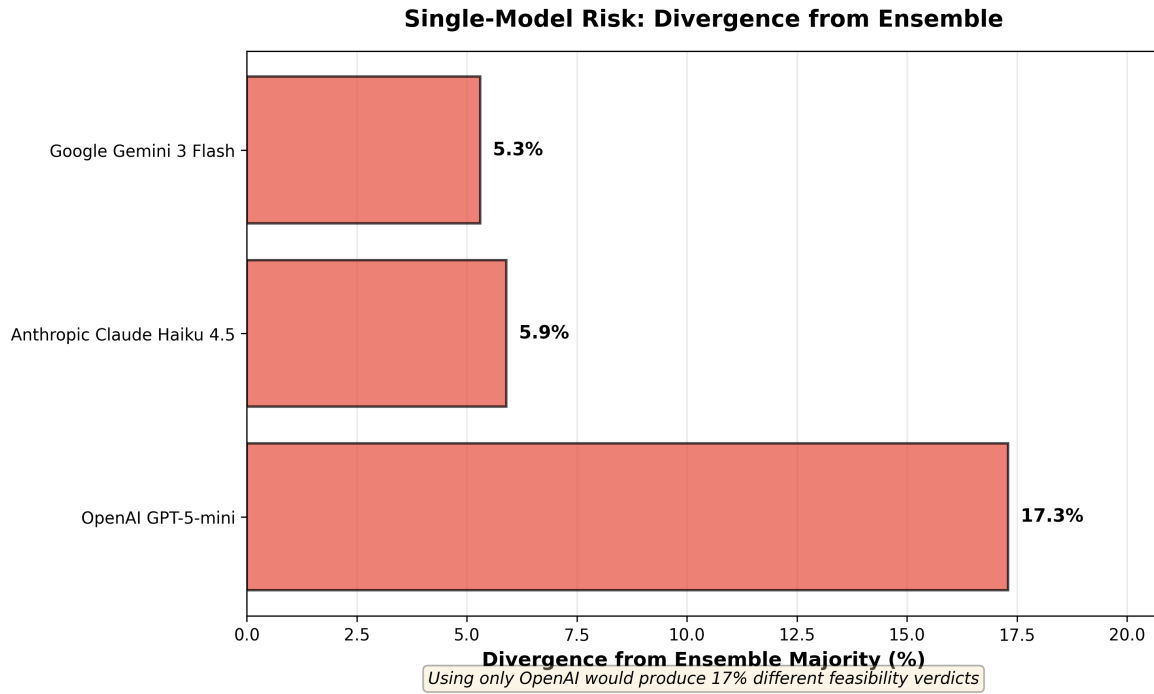


Arbitrator	Synthesis Rate	Style
Google (Gemini 3 Pro)	7%	Deferential — trusts raters
OpenAI (GPT-5.2)	59%	Balanced
Anthropic (Claude Opus 4.5)	77%	Active synthesis

No single model dominates. Ensemble balances conservative and synthetic approaches.

Why Not Just Use One Model?

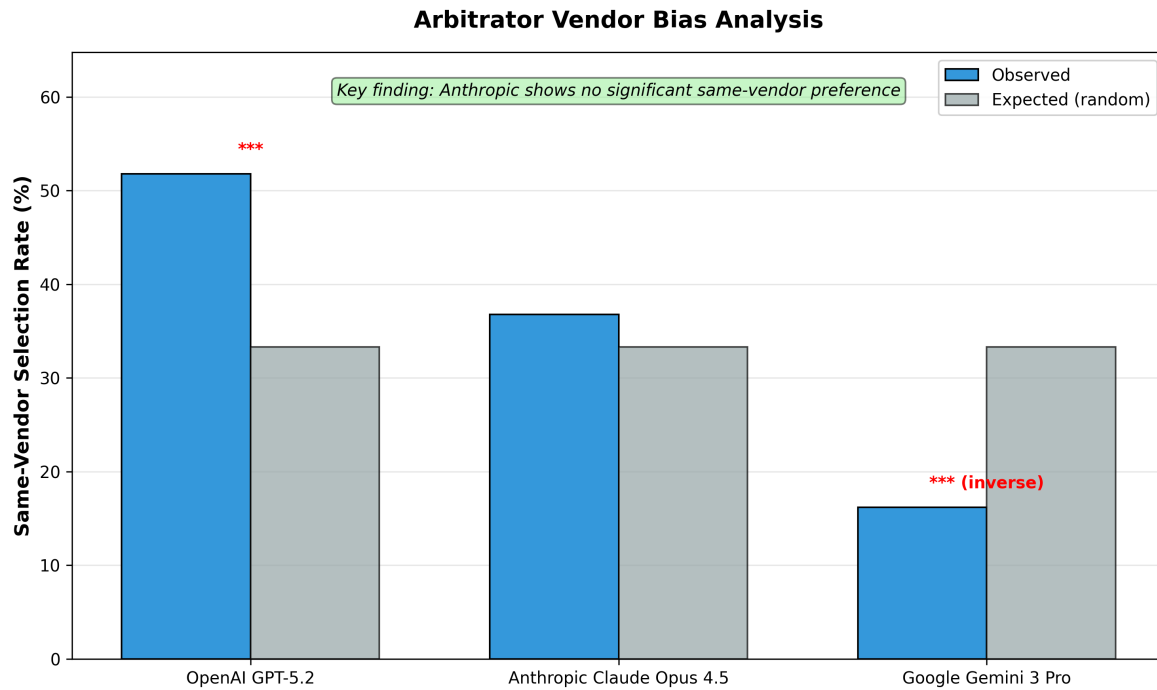
Single-model divergence from ensemble majority:



- **OpenAI alone:** 17.3% different feasibility verdicts (~277 pairs)
- **Anthropic alone:** 5.9% (~94 pairs)
- **Google alone:** 5.3% (~85 pairs)

The 3rd model catches what any 2 would miss.

Do Arbitrators Favor Their Own Vendor's Rater?



Arbitrator	Same-Vendor Selection	Expected	Significant?
OpenAI	51.8%	33.3%	Yes ($p < 0.001$)
Anthropic	36.8%	33.3%	No ($p = 0.159$)
Google	16.2%	33.3%	Yes — AGAINST own

Finding: Anthropic shows no same-vendor preference. Google actively avoids own vendor.

Cost/Quality: Is the 3rd Model Worth It?

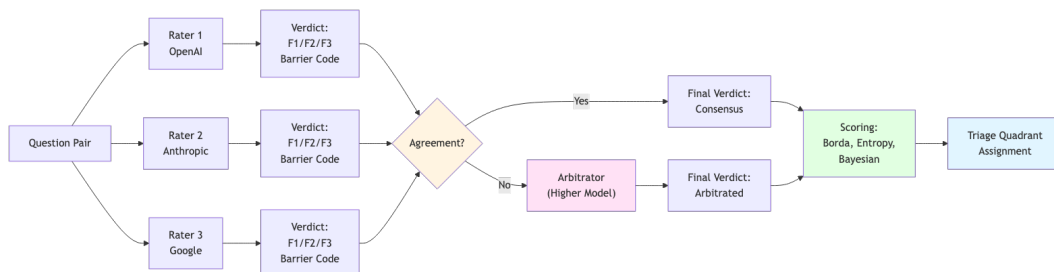
Two-model scenarios — tie frequency:

Pair	Feasibility Ties
OpenAI + Anthropic	23.2%
OpenAI + Google	22.1%
Anthropic + Google	11.0% ← lowest

Recommendation: - <1,000 pairs: 2 models + human tiebreaking - >1,000 pairs: 3 models for automated triage

The 3rd model's value is highest when first two disagree.

LLM Ensemble Pattern



Methodology Validation Summary

Validation Check	Result	Interpretation
Rater convergence	= 0.611	Substantial — task well-defined
Arbitrator convergence	= 0.843	Almost perfect — disagreements resolvable
Single-model risk	5-17%	Multi-model approach justified
Vendor bias	Mixed	Anthropic unbiased; OpenAI self-favoring

Conclusion: Multi-model ensemble provides robust classifications suitable for survey harmonization analysis.

Implications for AI-Assisted Research

- **Construct validity** through independent architecture convergence
- **Behavioral diversity** provides multiple valid perspectives
- **Cost/quality tradeoffs** enable scaling to larger datasets
- **Transparent biases** better than hidden single-model assumptions

Reusable methodology for future survey harmonization work

What's Next

1. Apply to additional federal surveys
 2. Extend to survey editing/imputation use cases
 3. Investigate optimal ensemble sizes for different task types
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Summary

Research Question: Can multi-model LLM ensembles reliably classify survey harmonization?

Answer: Yes, with documented construct validity

Evidence: - High convergence ($\rho = 0.611 \rightarrow 0.843$) - Documented behavioral differences - Quantified single-model risk - Vendor bias patterns identified

Impact: Validated methodology for AI-assisted survey analysis

For survey-specific findings and burden reduction implications, see Report 3A

Questions?

Contact: [your info]

Repository: [link]

Methodology documentation: [link]

Appendix

Harmonization Framework

Adopted standard survey harmonization classification:

Level	Definition
F1	Directly consolidable — no transformation needed
F2	Consolidable with transformation — needs recoding
F3	Not consolidable — fundamental barriers

Barrier codes: CC (construct/concept), TC (temporal), RS (response scale)

Harmonization Taxonomy

Primary Source: Fortier et al. (2011, 2017) DataSHaPER/Maelstrom framework

Supporting: Wolf et al. (2016), Saris & Gallhofer (2014), Slomczynski & Tomescu-Dubrow (2018)

Code	Feasibility	Definition	Action Required
F1	Direct recode	Mechanically transformable with simple recoding or collapsing	Simple data transformation
F2	Statistical adjustment	Requires modeling, imputation, or bridging studies	Statistical harmonization
F3	Incompatible	Fundamentally different constructs; not harmonizable without re-fielding	No harmonization possible

Fortier et al. (2011) found 38% of variables could be harmonized across 53 studies

Barrier Code Taxonomy (1/3)

Temporal Constraints (TC)

Code	Subtype	Definition	Example
TC	Temporal/Chronological	Reference period or timing differences	—
TC.1	Reference period length	Different recall windows	7-day vs 12-month
TC.2	Temporal framing	Point-in-time vs habitual vs retrospective	“last week” vs “usually”
TC.3	Calendar alignment	Fixed vs rolling reference periods	Calendar year vs “past 12 months”

Construct Constraints (CC)

Code	Subtype	Definition	Example
CC	Construct/Concept	Concept definition or operationalization differences	—
CC.1	Concept definition	Different meaning of core term	“Employment” including/excluding unpaid work
CC.2	Operationalization	Different behavioral indicators	“Food insecurity” via different batteries
CC.3	Boundary conditions	Different thresholds or cutoffs	Income brackets with different boundaries
CC.4	Scope inclusions	Different components counted	“Income” including/excluding specific sources

Barrier Code Taxonomy (2/3)

Population/Coverage Constraints (PC)

Code	Subtype	Definition	Example
PC	Population/Coverage	Universe, frame, or sample design differences	—
PC.1	Universe definition	Target population differs	Civilian vs all residents
PC.2	Frame exclusions	Different sampling exclusions	Group quarters, territories
PC.3	Age bounds	Different age eligibility	15+ vs 16+ vs 18+
PC.4	Geographic scope	Different geographic coverage	50 states vs states + territories

Response Scale Constraints (RS)

Code	Subtype	Definition	Example
RS	Response Scale	Scale type, categories, or format differences	—
RS.1	Scale type	Fundamentally different formats	Binary vs 5-point vs continuous
RS.2	Category structure	Different number/boundaries	5-point vs 7-point Likert
RS.3	Anchoring/labels	Different verbal anchors	“Strongly agree” differences
RS.4	Numeric vs verbal	Number scale vs labeled categories	0-10 slider vs frequency labels

Barrier Code Taxonomy (3/3)

Mode/Context Constraints (MC)

Code	Subtype	Definition	Example
MC	Mode/Context	Interview mode or questionnaire context differences	—
MC.1	Interview mode	Different data collection modes	CATI vs web self-administered

Code	Subtype	Definition	Example
MC.2	Question routing	Different skip patterns	Subset vs all respondents
MC.3	Contextual priming	Preceding questions affect interpretation	Question order effects
MC.4	Proxy response	Proxy vs self-report rules	Household head vs individual

Processing/Metadata Constraints (PM)

Code	Subtype	Definition	Example
PM	Processing/Metadata	Coding, weighting, or documentation differences	—
PM.1	Coding schemes	Different classification systems	Occupation coding differences
PM.2	Derived variables	Different construction algorithms	Poverty calculation methods
PM.3	Documentation gaps	Insufficient metadata	Missing questionnaire context

Key Citations

Harmonization Framework:

- Fortier, I., et al. (2011). Is rigorous retrospective harmonization possible? Application of the DataSHaPER approach across 53 large studies. *International Journal of Epidemiology*, 40(5), 1314-1328.
- Fortier, I., et al. (2017). Maelstrom Research guidelines for rigorous retrospective data harmonization. *International Journal of Epidemiology*, 46(1), 103-115.

Survey Methodology:

- Wolf, C., et al. (2016). Harmonizing survey questions between cultures and over time. In *The SAGE Handbook of Survey Methodology* (pp. 502-524). SAGE.
- Saris, W. E., & Gallhofer, I. N. (2014). *Design, Evaluation, and Analysis of Questionnaires for Survey Research* (2nd ed.). Wiley.

Data Harmonization:

- Slomczynski, K. M., & Tomescu-Dubrow, I. (2018). Basic principles of survey data recycling. In *Advances in Comparative Survey Methods* (pp. 937-962). Wiley.
- Tomescu-Dubrow, I., et al. (Eds.). (2024). *Survey Data Harmonization in the Social Sciences*. Wiley.